IndicMMLU-Pro: Benchmarking Indic Large Language Models on Multi-Task Language Understanding

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Abstract

Indic languages, spoken by over 1.5 billion people, pose unique challenges for NLP due to their cultural richness, linguistic diversity, and structural complexity. We present **IndicMMLU-Pro**, a comprehensive benchmark extending the MMLU-Pro framework to nine major Indic languages: Hindi, Bengali, Gujarati, Marathi, Kannada, Punjabi, Tamil, Telugu, and Urdu. Covering a wide range of tasks in comprehension, reasoning, and generation, IndicMMLU-Pro offers a standardized evaluation framework to advance AI model development in Indic contexts. This paper details the benchmark's design, taxonomy, and data curation, and establishes baseline results using state-of-the-art multilingual models. As an open resource,[1] IndicMMLU-Pro aims to accelerate progress in Indic language technologies and support inclusive research in multilingual NLP.

1 Introduction

With over 1.5 billion speakers, Indic languages represent a significant portion of the world's linguistic diversity, reflecting the cultural and historical richness of the Indian subcontinent. Spanning the Indo-Aryan and Dravidian language families, these languages have evolved over centuries, playing a central role in everyday communication across domains such as education, governance, media, healthcare, and social services. Beyond their linguistic value, focused research and benchmarking on Indic languages can drive inclusive technological development—particularly in enhancing accessibility for the deaf and hard-of-hearing communities through improved support for Indic sign languages (Ananthanarayana et al., 2021b). Such efforts can foster greater social inclusion, broaden educational and employment access, and empower civic engagement among sign language users.

Despite their societal significance, Indic languages remain vastly underrepresented in Natural Language Processing (NLP) research. A comprehensive survey by KJ et al. (2024a) and findings from Joshi et al. (2020) highlight a pronounced imbalance in resource allocation, where only 28% of the world's languages are considered "winners" in terms of NLP attention, while the majority remain "left behind." This disparity is evident in the contrast between English and Bengali—two languages with comparable speaker bases—yet with English vastly outpacing Bengali in terms of digital resources, corpus availability, and research prominence (Lane, 2019).

This underrepresentation stems from several challenges unique to Indic languages, including complex morphological structures, syntactic variation, and a dearth of high-quality annotated datasets (Kakwani et al., 2020a; Marreddy et al., 2022). Additionally, historical and institutional biases have contributed to the colonial neglect of indigeneous languages in technology development (Das et al., 2024). However, growing interest in AI-driven applications and the transformative potential of NLP technologies for regional languages (Ananthanarayana et al., 2021a) underscore the pressing need for robust benchmarks to systematically evaluate and improve model performance in the Indic context.

To bridge this gap, we propose **IndicMMLU-Pro**, a comprehensive benchmark inspired by the recently released MMLU-Pro (Wang et al., 2024). IndicMMLU-Pro adapts the robust, multi-task evaluation framework of MMLU-Pro to the linguistic landscape of Indic languages, enabling the assessment of multilingual models across reasoning, linguistic understanding, and generative

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[1]https://huggingface.co/datasets/anonymous1069/IndicMMLU-Pro

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tasks. This paper makes three key contributions: (1) it introduces **IndicMMLU-Pro**, a comprehensive benchmark for evaluating AI models across diverse tasks and Indic languages; (2) it outlines the benchmark's design, taxonomy, and data curation, ensuring cultural and linguistic fidelity; and (3) it presents baseline results using state-of-the-art multilingual models, establishing a foundation for future Indic-language AI research.

2 Related Work

The development of multilingual language models has been fueled by the availability of diverse datasets, particularly for Indic languages, which pose unique challenges due to their rich morphology and diverse scripts. Early resources like WikiAnn (Pan et al., 2017) supported NER across many languages, including Indic ones. The IndicNLP Corpus (Kunchukuttan, 2020), IndicCorp (Joshi et al., 2022), and the IndicNLPSuite (Kakwani et al., 2020b) have collectively enabled significant progress in training models tailored to Indic languages, supporting tasks such as machine translation and classification.

To evaluate model performance, benchmarks like IndicGLUE (Kakwani et al., 2020c) have provided standardized tasks including sentiment analysis, natural language inference, and question answering across multiple Indic languages. Similarly, IndicXNLI (Aggarwal et al., 2022) targets inference tasks, while IndicNLG (Kumar et al., 2022) supports generation tasks such as translation and summarization, addressing linguistic diversity and syntactic complexity.

Beyond the Indic landscape, Megaverse (Ahuja et al., 2023) and MultiQ (Holtermann et al., 2024) assess large language models across multiple languages and modalities, with an emphasis on foundational multilingual capabilities and performance in underrepresented languages.

Finally, general-purpose multilingual benchmarks like MMLU (Liang et al., 2020; Hu et al., 2020) have set a high bar for evaluating language understanding across a wide spectrum of tasks, offering key insights into the strengths and limitations of current multilingual models.

3 Methodology

To create IndicMMLU-Pro, a benchmark for Indic languages equivalent to MMLU Pro, we adopted a process similar to the prior work on IndicMMLU, organizing our approach into two main steps: dataset creation and baseline benchmarking.

3.1 Dataset Creation

Our goal was to provide MMLU-Pro in nine Indic languages: Hindi, Bengali, Telugu, Marathi, Tamil, Gujarati, Urdu, Kannada and Punjabi. To achieve this, we used IndicTrans2 (Gala et al., 2023), a state-of-the-art machine translation model specifically designed for Indic languages.

As shown in Figure1, we use IndicTrans2 to convert the questions and corresponding options from the original English MMLU Pro dataset into each of the target languages. This approach allowed us to maintain the structure and content of the original benchmark while adapting it to the linguistic characteristics of Indic languages. The experimental settings for IndicTrans2 were as follows: *Model Size*-1B parameters, *Quantization*-None, *Batch Size*-8. We applied these settings consistently across all nine language translations to ensure uniformity in the translation process.

3.2 Quality Assurance

To maintain the integrity and accuracy of the translated content, we implemented a rigorous quality assurance process:

Back-translation: For a subset of the data, we performed back-translation to English and compared it with the original text to identify any significant discrepancies.

Validation: After back-translation, we verified the back-translated dataset with the original MMLU-Pro on numerous metrics, such as chrF++ BLEU, METEOR, TER & SacreBLEU to ensure the dataset's quality and consistency, also described in Section 4. This multi-metric evaluation provides a comprehensive assessment of the translation's accuracy and fluency.

3.3 Dataset Structure

The resulting IndicMMLU-Pro dataset maintains the same structure as the original MMLU-Pro dataset, with separate subsets for each of the nine Indic languages. The dataset retains the original categories and task types from MMLU Pro, thus ensuring identical usage and facilitating comprehensive evaluation in various domains and cognitive skills.

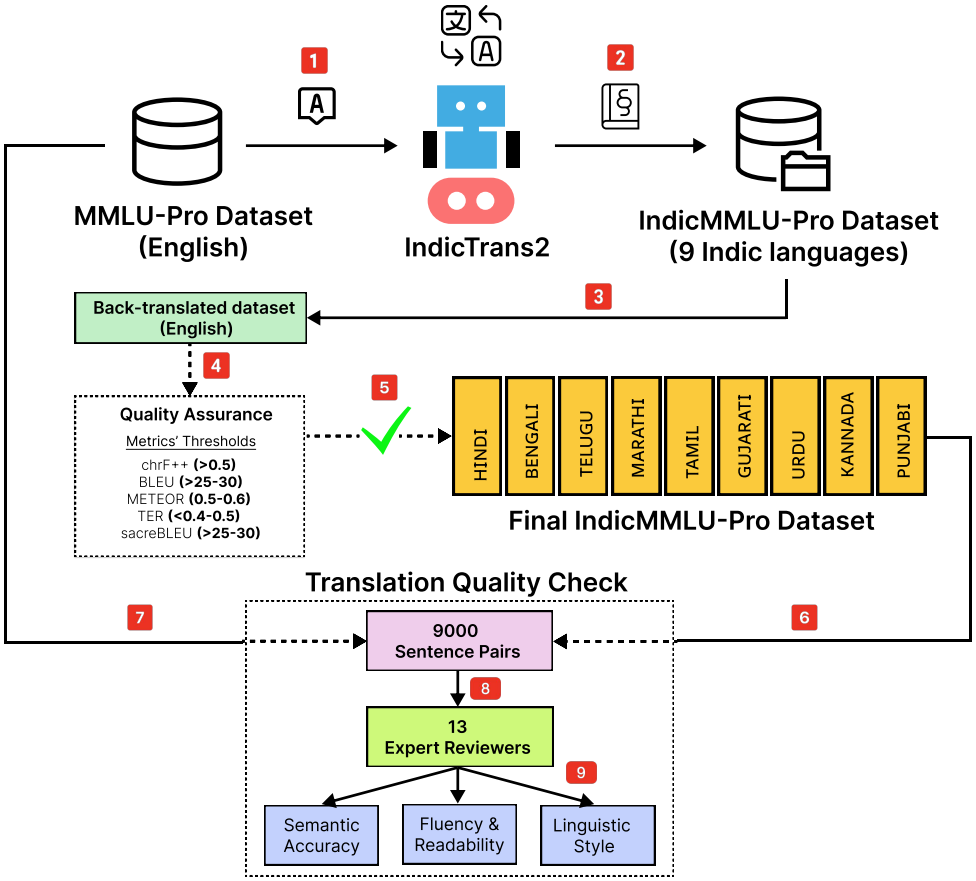


Figure 1: IndicMMLU-Pro Dataset Construction and Evaluation Pipeline. The diagram illustrates the end-to-end process of creating and validating the IndicMMLU-Pro dataset across nine Indic languages. Starting with the English MMLU-Pro dataset, content is translated using IndicTrans2 (1B parameters) and undergoes rigorous quality assurance through back-translation and multiple metric evaluations (chrF++, BLEU, METEOR, TER, and SacreBLEU). Only translations meeting quality thresholds proceed to the final dataset. The workflow also shows the comprehensive evaluation process including expert proofreading involving 13 reviewers who assess semantic accuracy, fluency, and linguistic style. This systematic approach ensures the creation of a high-quality, multilingual benchmark dataset that maintains the integrity of the original MMLU-Pro while adapting to the linguistic nuances of Indic languages.

3.4 Dataset Availability

The IndicMMLU-Pro dataset is publicly available on the Hugging Face Hub (KJ et al., 2024b). This allows for easy access and reproducibility of our results. Researchers and practitioners can directly use or adapt this dataset for their studies and applications in the processing of Indic languages.

3.5 Baseline Benchmarking

To establish baseline performance metrics for the IndicMMLU-Pro benchmark, we evaluated state-of-the-art multilingual language models: GPT-4o, GPT-4o-mini, Llama-3.1-8B-Instruct, IndicBERT (Kakwani et al., 2020a), IndicBART (Dabre et al., 2021), RemBERT (Chung et al., 2020), MuRIL (Khanuja et al., 2021), and XLM-RoBERTa (Conneau et al., 2020), Navarasa, Airavata (Gala et al., 2024), OpenHathi, TamilLlama (Balachandran, 2023), and MahaMarathi. These models were chosen due to their demonstrated capabilities in handling multiple languages and their specific design for the inclusion of Indic languages in their training data. The benchmarking process was conducted as follows:

Model Selection: To establish baseline performance across the nine Indic languages in IndicMMLU-Pro, we evaluate a diverse set of language models with demonstrated capabilities in Indic NLP.

Data Preparation: The *test* split for each language was used, with data preprocessed to conform to the input requirements of each model.

Evaluation: Accuracy, defined as the percent-

age of correct predictions, serves as the primary metric. While we acknowledge the importance of complementary evaluation measures for capturing diverse performance aspects (Bhaduri et al., 2024a; Bedemariam et al., 2025), accuracy provides a suitable baseline. Evaluation was conducted separately for each language to enable granular, language-specific analysis.

Computational Resources: All experiments were run on a cluster of NVIDIA A100 GPUs, with each model evaluation averaging 24 hours per language.

The benchmarking results, presented in Table 1, provide a comprehensive overview of model performance across languages and serve as a foundation for future advancements in Indic-language NLP.

3.6 Proofreading and Setup

We conducted a large-scale proofreading exercise involving 9,000 pairs of sentences (English-Indic), comprising 1,000 pairs of test sets from each of the nine languages in the IndicMMLU-Pro dataset. These pairs were stratified into 14 categories and 4 data sources per language, then randomly shuffled across all Indic languages to eliminate source- or category-based biases. The proofreading process was guided by three key evaluation criteria:

- **Semantic Accuracy and Correctness**: The translation must preserve the exact meaning of the source without introducing errors, omissions, or additions.
- **Fluency and Readability**: The translated sentence should read naturally and be easily comprehensible in the target language.
- **Linguistic and Stylistic Appropriateness**: The tone, style, and usage should be appropriate for the intended audience and cultural context.

Thirteen experts participated in this effort, representing all nine Indic languages. Each language had at least one native expert, while four languages had two experts to balance workload. All experts were fluent in English and had formal education in their respective native languages.

Each sentence pair was evaluated using a standardized 5-point scale (1 = lowest, 5 = highest) across the three criteria. To ensure uniformity, comprehensive guidelines, and illustrative examples were provided in advance. Expert evaluations were systematically recorded in spreadsheets, capturing scores and qualitative assessments. We gratefully acknowledge the contributions of all language experts (Section 6) who volunteered their time and expertise to support this research.

4 Results

Table 3 presents the evaluation metrics for the IndicMMLU-Pro dataset using back-translation techniques. The chrF++ (Popović, 2017) and BLEU (Papineni et al., 2002) scores are provided for each of the nine Indic languages (Bengali, Gujarati, Hindi, Kannada, Marathi, Punjabi, Tamil, Telugu, and Urdu). These metrics assess the quality and accuracy of the translated content in the IndicMMLU-Pro benchmark, providing insights into the dataset's linguistic fidelity across different Indic languages.

Table 1 and 2 present the accuracy scores (in percentages) of various pre-trained language models evaluated on the IndicMMLU-Pro benchmark. The benchmark covers nine major Indic languages: Bengali, Gujarati, Hindi, Kannada, Marathi, Punjabi, Tamil, Telugu, and Urdu. The scores reflect the models' ability to handle diverse linguistic challenges specific to Indic languages.

4.1 Overall Performance

Performance Range: The accuracy scores across all models and languages now fall within a broader range of 9.90% to 44.80%. This wider range reflects the significant performance differences between the newer, more advanced models (like GPT-4o) and the previously evaluated models.

Model Comparison: GPT-4o consistently outperforms all other models across all languages, achieving the highest scores ranging from 38.46% to 44.80%. GPT-4o mini follows as the second-best performer with scores ranging from 25.75% to 35.08%. Among the previously evaluated models, XLM-RoBERTa remains the top performer, consistently outperforming other models across most languages with scores ranging from 11.92% to 13.16%.

Language-Model Variability: There is now a much more pronounced variability in performance across different languages and models. This suggests that language-specific characteristics and

Language	GPT-4o	GPT-4o mini	Llama-3.1-8B	IndicBART	IndicBERT	RemBERT	MuRIL
Hindi	**44.80**	32.33	18.61	11.21	10.78	11.41	10.87
Bengali	**44.38**	31.11	N/A	12.52	10.39	12.00	9.90
Punjabi	**40.60**	26.25	N/A	11.78	10.36	11.06	10.36
Marathi	**42.20**	27.13	N/A	11.65	10.59	12.93	11.79
Urdu	**44.18**	31.13	N/A	12.11	11.63	11.32	11.20
Gujarati	**41.77**	28.29	N/A	12.14	11.06	12.13	10.79
Telugu	**41.34**	26.78	N/A	12.05	11.36	10.20	9.96
Tamil	**38.46**	35.08	N/A	11.70	10.96	10.98	11.00
Kannada	**38.97**	25.75	N/A	11.51	11.71	10.87	10.62

Table 1: Performance comparison of language models on the IndicMMLU-Pro benchmark across nine Indic languages, including Indo-Aryan (Hindi, Bengali, Punjabi, Marathi, Urdu, and Gujarati) and Dravidian (Telugu, Tamil, and Kannada) languages. Accuracy scores are shown as percentages. Models compared include GPT-4o, GPT-4o mini, IndicBART, IndicBERT, RemBERT, MuRIL, and Llama-3.1-8B-Instruct.

Language	XLM-RoBERTa	Navarasa	Airavata	OpenHathi	TamilLlama	MahaMarathi
Hindi	12.33	12.43	11.60	11.65	-	-
Bengali	12.68	12.08	-	-	-	-
Punjabi	12.59	11.95	-	-	-	-
Marathi	12.57	11.88	-	-	-	11.60
Urdu	12.53	10.73	-	-	-	-
Gujarati	11.92	11.53	-	-	-	-
Telugu	12.62	11.77	-	-	11.53	-
Tamil	12.34	12.38	-	-	11.66	-
Kannada	13.16	11.88	-	-	-	-

Table 2: Comparison of language model performance across Indian languages, both Indo-Aryan (i.e., Hindi, Bengali, Punjabi, Marathi, Urdu, and Gujarati) and Dravidan (i.e., Telegu, Tamil, and Kannada). Scores are shown for Llama 3.1, Navarasa, Airavata, OpenHathi, TamilLlama, and MahaMarathi models where available.

model architectures play significant roles in performance outcomes.

4.2 Language-wise Performance

Breaking down the performance for each language reveals interesting patterns:

Hindi: GPT-4o leads with 44.80%, followed by GPT-4o mini (32.33%). Among other models, Navarasa (12.43%) and XLM-RoBERTa (12.33%) perform best.

Bengali: GPT-4o achieves 44.38%, with GPT-4o mini at 31.11%. XLM-RoBERTa (12.68%) and Navarasa (12.08%) lead among other models.

Telugu: GPT-4o scores 41.34%, GPT-4o mini 26.78%. XLM-RoBERTa (12.62%) outperforms Navarasa (11.77%) and TamilLlama (11.53%).

Marathi: GPT-4o reaches 42.20%, GPT-4o mini 27.13%. RemBERT (12.93%) performs best among earlier models, with Navarasa at 11.88% and MahaMarathi at 11.60%.

Tamil: GPT-4o mini shows strong performance at 35.08%, close to GPT-4o's 38.46%. Navarasa (12.38%) slightly outperforms XLM-RoBERTa (12.34%), with TamilLlama at 11.54%.

Gujarati: GPT-4o scores 41.77%, GPT-4o mini 28.29%. IndicBART (12.14%) and RemBERT (12.13%) perform well among earlier models.

Urdu: GPT-4o achieves 44.18%, GPT-4o mini 31.13%. XLM-RoBERTa (12.53%) leads among other models.

Kannada: GPT-4o scores 38.97%, GPT-4o mini 25.75%. XLM-RoBERTa significantly outperforms other models (13.16%) among earlier models.

Punjabi: GPT-4o reaches 40.60%, GPT-4o mini 26.25%. XLM-RoBERTa (12.59%) performs best among other models.

4.2.1 Cross-linguistic Analysis

IndicMMLU-Pro reveals distinct performance trends across Indo-Aryan (Hindi, Bengali, Punjabi, Gujarati, Urdu, Marathi) and Dravidian (Kannada, Tamil, Telugu) languages.

Performance Trends: GPT-4o and GPT-4o mini consistently outperform all other models across both language families. Among smaller mod-

els, XLM-RoBERTa shows strong overall performance, while Navarasa is particularly competitive on Dravidian languages.

Language-specific Models: TamilLlama and MahaMarathi perform well within their respective domains but fall short of top multilingual baselines such as XLM-RoBERTa and Navarasa.

Consistency: GPT-4o variants exhibit robust, uniform performance across languages, with Navarasa also showing strong consistency, especially among Dravidian languages.

Script Effects: Languages sharing the Devanagari script (e.g., Hindi, Marathi) display similar trends, while Urdu, written in the Perso-Arabic script, shows greater variability.

Model Gap: The substantial gap between GPT-4o models and others underscores the advantage of larger, instruction-tuned models in processing linguistically diverse Indic languages.

4.3 Dataset Quality Assessment

To heuristically assess the quality of the IndicMMLU-Pro dataset, we perform back-translation using IndicTrans2 (see Section 3.1), converting Indic-language test samples back to English. The resulting English translations are evaluated using standard translation metrics:

chrF++ (Popović, 2017) (*threshold > 50%*): Character n-gram F1 score (with $n = 6$), agnostic to language and tokenization.

BLEU (Papineni et al., 2002) (*threshold > 25–30%*): Precision-based metric comparing n-grams of candidate and reference translations.

METEOR (Banerjee and Lavie, 2005) (*threshold = 50–60%*): Considers unigram precision and recall with synonymy and stemming.

TER (Snover et al., 2006) (*threshold < 40–50%*): Measures the number of edits needed to match the reference.

SacreBLEU (Post, 2018) (*threshold > 25–30%*): Standardized BLEU variant with consistent tokenization and processing.

Table 3 presents back-translation scores for Hindi, Gujarati, and Tamil. Hindi shows the highest quality across all metrics (chrF++: 78.06, BLEU: 0.59, METEOR: 0.56, TER: 42.27, SacreBLEU: 59.07), followed closely by Gujarati. Tamil performs slightly lower (e.g., chrF++: 74.32, TER: 46.41), though still within acceptable quality ranges.

High chrF++ scores (above 74) across all three languages indicate good semantic preservation. BLEU, METEOR, and SacreBLEU further support the quality of the translations, while TER scores suggest a moderate level of required post-editing, with Tamil requiring slightly more effort.

It should be noted that similar assessments for the remaining six languages, Bengali, Punjabi, Kannada, Telugu, Urdu, and Marathi, are currently unavailable. This limits the scope of our dataset quality analysis and underscores the need for further evaluation to ensure consistency across all supported languages.

In summary, back-translation metrics affirm high-quality translations for Hindi, Gujarati, and Tamil, with Hindi demonstrating the most robust performance. A more complete quality assessment will be necessary for a holistic evaluation of IndicMMLU-Pro. To better illustrate the meaning of chrF++ scores in practice, let's examine sample translations for Hindi (highest score), Gujarati (middle score), and Tamil (lowest score) as shown in Figure 2, 3, 4 respectively.

Hindi (chrF++: 78.06): The chrF++ score for Hindi is the best among the three languages indicating a high structural and semantic similarity between the original text and the back-translated text.

Gujarati (chrF++: 77.67): While slightly lower, this score still represents a high-quality translation with minor variations.

Tamil (chrF++: 74.32): This lower score might reflect slight changes in word choice or structure, but the core meaning is preserved.

These examples demonstrate that even with the lowest chrF++ score in our dataset (74.32 for Tamil), the translations maintain good semantic fidelity. The differences in scores often reflect nuances in word choice, sentence structure, or slight variations in conveying the same meaning, rather than significant errors in translation. This analysis supports our earlier observation that the IndicMMLU-Pro dataset maintains good overall translation quality across the evaluated languages.

Preliminary assessments of the other languages (Bengali, Punjabi, Kannada, Telugu, Urdu, and Marathi) suggest similar trends in translation quality, with chrF++ scores ranging from 73 to 79.

Language	chrF++	BLEU	METEOR	TER	SacreBLEU
Hindi	78.06	0.59	0.56	42.27	59.07
Gujarati	77.67	0.58	0.55	43.09	58.28
Tamil	74.32	0.54	0.52	46.41	53.64

Table 3: Back-translation evaluation metrics for the IndicMMLU-Pro dataset for 3 Indic languages.

Complete metrics for these languages are being compiled and will be included in future publications.

4.4 Cosine Similarity Scores

Language	Questions	Choices
Hindi	0.9109	0.9250
Bengali	0.9172	0.9251
Telugu	0.9193	0.9287
Marathi	0.9126	0.9242
Tamil	0.9194	0.9255
Gujarati	0.9164	0.9320
Urdu	0.9121	0.9302
Kannada	0.9149	0.9238
Punjabi	0.9177	0.9254

Table 4: Cosine similarity scores between LaBSE embeddings of IndicMMLU-Pro languages and English MMLU-Pro for questions and multiple-choice options. These scores are used as a measure of semantic similarity, with higher values suggesting closer meaning alignment across languages.

We assessed the semantic alignment between IndicMMLU-Pro and its English counterpart, MMLU-Pro, using the LaBSE (Language-agnostic BERT Sentence Embedding) model. For each language, we generated embeddings for corresponding questions and multiple-choice options from both datasets. The cosine similarity was then calculated to evaluate semantic closeness, yielding two key metrics: **average question similarity** and **average choice similarity**. This analysis provides insight into how well Indic translations preserve the meaning and structure of the original English content, an essential factor to ensure the quality and consistency of multilingual evaluation benchmarks in NLP.

5 Key Findings and Implications of IndicMMLU-Pro

IndicMMLU-Pro marks a significant advancement in evaluating AI models for Indic languages. Through comprehensive benchmarking across nine languages and multiple tasks, it offers valuable insights into the current capabilities and limitations of state-of-the-art multilingual models in handling the linguistic diversity of the Indian subcontinent. This section summarizes the key findings and their broader implications.

5.1 Language-Specific Insights

Performance Variability: Model performance varied significantly across languages, highlighting the need for language-specific adaptations in training and evaluation.

Script and Language Family Effects: Languages using the same script (e.g., Devanagari for Hindi and Marathi) showed similar performance trends, while Urdu (written in the Perso-Arabic script) exhibited more variance. Additionally, some models displayed greater consistency in Dravidian languages compared to Indo-Aryan ones, suggesting architectural sensitivities to linguistic typologies.

5.2 Dataset and Evaluation Quality

Translation Quality: Back-translation metrics (e.g., chrF++ > 74 for Hindi, Gujarati, Tamil) confirmed good semantic preservation. However, score discrepancies across metrics (e.g., BLEU vs. chrF++) revealed variability in phrasing and surface structure.

Incomplete Coverage: Only three languages were assessed for translation quality, underscoring the need for comprehensive evaluation across all nine languages.

5.3 Methodological Insights

Benchmark Design: IndicMMLU-Pro successfully adapts the MMLU-Pro framework to the Indic context, offering a scalable template for language-specific, multi-task benchmarks.

Data Creation Pipeline: The integration of IndicTrans2 with rigorous quality assurance ensures high-quality dataset generation, providing a replicable approach for future multilingual benchmark creation.

Multi-Metric Evaluation Employing diverse metrics (chrF++, BLEU, METEOR, TER, SacreBLEU) allowed for a nuanced assessment of

Original	Hindi	Back-translated
A total of 30 players will play basketball at a park. There will be exactly 5 players on each team. Which statement correctly explains how to find the number of teams needed?	एक पार्क में कुल 30 खिलाड़ी बास्केटबॉल खेलेंगे प्रत्येक टीम में ठीक 5 खिलाड़ी होंगे कौन सा कथन सही ढंग से बताता है कि आवश्यक टीमों की संख्या कैसे ज्ञात की जाए?	A total of 30 players will play basketball in a park. Each team will have exactly 5 players. Which statement correctly states how to find the number of teams required?

Figure 2: The original text sample, its Hindi translation, and the corresponding back-translated text

Original	Gujarati	Back-translated
Which of the following is the body cavity that contains the pituitary gland?	નીચેનામાંથી કઈ શરીરની પોલાણ છે જેમાં કફોત્પાદક ગ્રંથિ હોય છે?	Which of the following is the body cavity that contains the pituitary gland?

Figure 3: The original text sample, its Gujarati translation, and the corresponding back-translated text

Original	Tamil	Back-translated
What is the embryological origin of the hyoid bone?	ஹையாய்டு எலும்பின் கருப்பொருள் தோற்றம் என்ன?	What is the thematic appearance of the hyoid bone?

Figure 4: The original text sample, its Tamil translation, and the corresponding back-translated text

translation and semantic fidelity, capturing multiple dimensions of dataset quality.

These findings highlight the potential of IndicMMLU-Pro as both an evaluation tool and a guide for advancing model performance across Indic languages.

5.4 Future Directions

Based on our findings, we identify the following priorities for future research:

Data Collection: Expand high-quality datasets for low-resource Indic languages to support more equitable model training and evaluation.

Model Development: Design architectures and pretraining strategies that better capture the morphological and syntactic richness of Indic languages.

Cross-lingual Transfer: Enhance knowledge transfer between related Indic languages to improve low-resource performance via multilingual or linguistically-informed training.

Task-Specific Fine-tuning: Explore optimal fine-tuning techniques and Indic-centric pretraining tasks to boost model specialization (Balne et al., 2024).

Evaluation Metrics: Refine metrics to account for linguistic and cultural nuances often overlooked by existing evaluation frameworks.

IndicMMLU-Pro sets a new benchmark for Indic-language evaluation, enabling researchers and practitioners to assess, compare, and improve multilingual models in culturally and linguistically diverse contexts. By fostering progress in model development, task generalization, and language inclusion, it provides a roadmap for more accessible and representative AI systems.

To advance this agenda, interdisciplinary collaboration is essential. Incorporating insights from linguistics, social sciences, and cultural studies can guide the development of models that are not only technically robust but also socially informed (Bhaduri et al., 2024b; Mackenzie et al., 2024). Such approaches will help bridge the gap between language technologies and the communities they aim to serve.

IndicMMLU-Pro offers a robust, scalable framework for evaluating multilingual models across Indic languages. Although challenges remain, particularly in data coverage and equitable language representation, this benchmark provides a foundation for progress. As the field of NLP continues to evolve, tools like IndicMMLU-Pro will be crucial for ensuring that technological advancements reach and benefit all language communities, fostering inclusivity and accessibility in the AI era.

6 Acknowledgements

We extend our deepest gratitude to the experts who volunteered their time and effort to evaluate our dataset. Unfortunately, Indic languages are often neglected in the development and evaluation of Large Language Models, making initiatives like ours reliant on the generosity and expertise of individuals like those listed below. The list of experts has been kept anonymous for the double-blind re-

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